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Nomenclature

Physical constants

ε_0	Permittivity of free space
μ_0	Permeability of free space
c	Speed of light in vaccuum
e	Elementary charge
h	Planck constant
k_B	Boltzmann constant
m_e	Electron mass
N_A	Avogadro's number

Scalars

α	Attenuation coefficient	m^{-1}
Γ	Reaction rate density	$\mathrm{m}^{-3}\,\mathrm{s}^{-1}$
ε_r	Relative permittivity	1
η	Refractive index	1
μ	Dynamic viscosity	$\mathrm{Pa\,s}$
μ_r	Relative permeability	1
ν	Frequency	Hz
ρ	Charge density	$\mathrm{C\,m}^{-3}$
ρ	Mass density	$\mathrm{kg\,m}^{-3}$
σ	Reaction cross section	m^2
Φ	Quantum yield	1
ϕ	Scalar flux	$\mathrm{m}^{-2}\,\mathrm{s}^{-1}$

ω	Angular frequency	$\mathrm{rad\,s^{-1}}$
A	Area	$\mathrm{m^2}$
$\mathcal{E}$	Particle energy	J
ρE	Energy density	$\mathrm{J\,m^{-3}}$
E	Specific energy	$\mathrm{J\,kg^{-1}}$
$\mathcal{H}$	Particle enthalpy	J
ρH	Enthalpy density	$\mathrm{J\,m^{-3}}$
H	Specific enthalpy	$\mathrm{J\,kg^{-1}}$
I	Intensity	$\mathrm{W\,m^{-2}}$
k	Thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
M	Molar mass	$\mathrm{kg\,mol^{-1}}$
m	Mass	kg
N	Number	1
n	Number density	$\mathrm{m^{-3}}$
p	Pressure	Pa
q'''	Volumetric heat generation rate	$\mathrm{W\,m^{-3}}$
S	Reaction rate coefficient	varies
s	Distance	m
T	Temperature	K
t	Time	s
V	Volume	$\mathrm{m^3}$
v	Speed	$\mathrm{m\,s^{-1}}$
Vectors		
$\hat{\boldsymbol{\omega}}$	Direction	1
$\mathbf{a}$	Acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{B}$	Magnetic field	T
$\mathbf{E}$	Electric field	$\mathrm{V\,m^{-1}}$
$\mathbf{f}$	Force density	$\mathrm{N\,m^{-3}}$

$\mathbf{g}$	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{j}$	Current density	$\mathrm{A\,m^{-2}}$
$\hat{\mathbf{n}}$	Normal direction	1
$\mathbf{q}''$	Heat flux density	$\mathrm{W\,m^{-2}}$
$\mathbf{r}$	Position	m
$\mathbf{S}$	Poynting vector	$\mathrm{W\,m^{-2}}$
$\mathbf{u}$	Velocity	$\mathrm{m\,s^{-1}}$
$\rho\mathbf{u}$	Momentum density	$\mathrm{N\,s\,m^{-3}}$

Matrices

$\mathbf{T}$	Deviatoric stress tensor	Pa

Subscripts

ci	Collisional ionization
exc	Excitation
ib	Inverse Bremsstrahlung
pi	Photoionization
rec	Recombination
b	Bound electron
e	Electron
ei	Electron-ion interactions
en	Electron-neutral interactions
i	Ion
in	Ion-neutral interactions

Chapter 1

Introduction

Chapter 2

Physics of laser-droplet interactions

Modeling even a single droplet that interacts with a laser beam is more complicated than a typical thermal-hydraulic simulation. Even as the laser cross section is pretty small compared to the droplet surface area, the migration of free charged particles necesitates treating the entire droplet as a plasma. Modeling of plasmas, especially weakly-ionized ones, typically enforces strong coupling between the conservation equations and Maxwell's equations.

Chemically speaking, the laser can depart enough energies to not only remove bound electrons from their orbits but also break molecular bonds and alter the chemical composition of the fluid. In fact, free radicals and light molecular ions are frequently encountered in plasmas if the laser is not strong enough to completely disintegrate molecules into monoatomic ions. However, for simplicity, all ions are assumed to have resulted from neutral molecules each losing a single electron, and thus they have a charge of +1 and the same molecular weight as the parent neutral particle. With three separate species - electrons, ions, and neutral particles - in the system, a three-species model is developed. Under this approach, electrons, ions, and neutral particles are treated as separate fluids, each having its own set of state variables and interacting with the laser pulse, the electric and the magnetic fields in its own way. As the species try to equilibrate, their interactions with one another prove equally important.

2.1 Laser interactions

2.1.1 Energy deposition

This model assumes the laser beam is a line heat source within each droplet even if the laser cross-sectional area is known. The volumetric heat generation, as a result, contains a Dirac delta function $\delta(\boldsymbol{r}_{\text{laser}})$ where $\boldsymbol{r}_{\text{laser}}$ is the laser trajectory. Finite-volume methods blend well with this assumption since the delta function is always integrated out during the formulation.

$$\iiint_{\Gamma \supset \boldsymbol{r}_0} \delta(\boldsymbol{r}_0) dV = 1 \tag{2.1}$$

Therefore, the total energy deposited in a cell that the laser passes through is the difference in laser energy at the cell entrance and exit. If the entrance energy upon cell k is $I_k A$, it will have been attenuated by a factor of $e^{-\alpha_k s_k}$ when exitting the cell – s_k being the distance that the laser beam travels within cell k. Therefore,

$$\iiint_{V_k} q'''_{\text{laser}} dV = I_k A (1 - e^{-\alpha_k s_k}) \tag{2.2}$$

Laser energy absorption is done by electrons. Whereas the term "electrons" typically refers to free electrons for much of this chapter, in this context it collectively refers to both free and bound electrons, but the two kinds differ in how the energy is spent. A free electron, upon absorbing a photon, simply increases in kinetic energy. A bound electron, however, can either be excited or removed from its orbit altogether. If excitation occurs, collisions with the atom causes the electron to rapidly relax and transfer the acquired energy into thermal energy of the bulk liquid; therefore, on the timescale of interest, it is responsible to assume that the energy goes directly into heating the neutral fluid.

With two channels in which laser energy can be deposited, the total attenuation coefficient α is the sum of the bound and free counterparts.

$$\alpha = \alpha_b + \alpha_{\mathrm{ib}}, \tag{2.3}$$

and the fraction of energy that goes into each channel is proportional to the relative magnitude of the two absorption coefficients

$$q_b = I_0 A(1 - e^{-\alpha s})\frac{\alpha_b}{\alpha} \tag{2.4a}$$

$$q_{\mathrm{ib}} = I_0 A(1 - e^{-\alpha s})\frac{\alpha_{\mathrm{ib}}}{\alpha}. \tag{2.4b}$$

The subscript "ib" in Equation 2.3 refers to inverse Bremsstrahlung, a process in which free electrons, upon colliding with ions, absorbs energy and starts to accelerate. It should take over as the dominant absorption process once a sufficiently opaque medium is formed. The attenuation coefficient will increase proportionally to the inverse Bremsstrahlung frequency ν_{ib}

$$\alpha_{\mathrm{ib}} \equiv \frac{\nu_{\mathrm{ib}}\eta}{c}. \tag{2.5}$$

ν_{ib} itself is given by

$$\nu_{\mathrm{ib}} = \frac{N_e}{N_c}\nu_{\mathrm{ei}}, \tag{2.6}$$

where N_c is the electron critical density and ν_{ei} is the electron-ion collision frequency.

$$N_c \equiv \frac{m_e \pi \omega^2}{e^2} \tag{2.7a}$$

$$\nu_{\mathrm{ei}} \equiv \frac{4}{3}\sqrt{\frac{2\pi}{m_e}}\frac{N_e Z e^4 \ln\Lambda}{(k_B T_e)^{3/2}} \tag{2.7b}$$

The average ionization number Z is assumed to be 1, and the Coulomb logarithm $\ln\Lambda$ is given by

$$\ln\Lambda \equiv \ln\left[\frac{3}{2Ze^3}\sqrt{\frac{(k_B T_e)^3}{\pi N_e}}\right]. \tag{2.8}$$

2.1.2 Photoionization

Photoionization can only occur if the photon energy, $h\nu$, exceeds the binding energy $\mathcal{E}_b$. This is simply the photoelectric effect, and it is typically caused by photons in the ultraviolet regime. For a quantum yield $\Phi \in [0, 1]$ that accounts for the number of photoelectrons produced per incident photon, the photoionization rate density is

$$\Gamma_{\mathrm{pi}} = \Phi\frac{q_b'''}{h\nu}, \tag{2.9}$$

where q_b''' is related to q_b in Equation 2.4a as

$$\iiint q_b''' dV = q_b. \tag{2.10}$$

A newly freed electron will carry an excess energy of $h\nu - \mathcal{E}_b$, which will be deposited towards the energy of all free electrons with a rate of

$$q_{\mathrm{pi}}''' = (h\nu - \mathcal{E}_b)\Gamma_{\mathrm{pi}}. \tag{2.11}$$

Any absorption that do not result in ionization will instead excite bound electrons and serve as a heat source for the neutral particles with a rate of

$$q_{\mathrm{exc}}''' = (1 - \Phi)q_b''' = q_b''' - h\nu\Gamma_{\mathrm{pi}}. \tag{2.12}$$

It is possible for a bound electron to simultaneously absorb multiple photons, a phenomenon known as multiphoton ionization [1]. This instead becomes a strongly nonlinear process, and the probability of multiple photons being absorbed by the same electron increases strongly with laser intensity. The minimum number of photons needed to completely remove an electron from its orbit is

$$N \equiv \left\lceil \frac{\mathcal{E}_b}{h\nu} \right\rceil. \tag{2.13}$$

The N-photon ionization rate is proportional to the N-th power of a particular measure of photon flux. In [2], that measure is the photon scalar flux, $\phi = \dfrac{I}{h\nu}$. However, since the laser intensity monotonic decreases with distance travelled, it is better to calculate the ionization rate using a path-averaged scalar flux:

$$\overline{\phi} = \frac{\overline{I}}{h\nu} \tag{2.14a}$$

$$= \frac{1}{h\nu s}\int_0^s I_0 e^{-\alpha s'} ds' \tag{2.14b}$$

$$= \frac{I_0}{h\nu}\frac{1 - e^{-\alpha s}}{\alpha s} \tag{2.14c}$$

The rate is then given by

$$\Gamma_{\mathrm{pi}} = \min\left(S_N \overline{\phi}^N n, \frac{q_b'''}{Nh\nu}\right) \tag{2.15}$$

Here, S_N is the N-photon ionization rate coefficient and n is the neutral particle number density. The min function is to ensure that the actual ionization rate does not exceed the photon deposition rate. In order for Γ_{pi} to have the correct unit, the unit of S_N is $\mathrm{m}^{2N}\ \mathrm{s}^{N-1}$. If Equation 2.15 were to be in terms of another unit (for example, laser intensity) instead, then the unit of S_N must be changed accordingly. The term N-photon ionization cross section and the symbol σ_N are often used in literature to describe this coefficient [1] [2]; this nomenclature is not adopted here in order to ensure that all cross sections have units of area.

The rate of energy deposited into the neutrals and electrons are

$$q_{\mathrm{pi}}''' = (Nh\nu - \mathcal{E}_b)\Gamma_{\mathrm{pi}} \tag{2.16a}$$

$$q_{\mathrm{exc}}''' = q_b - Nh\nu\Gamma_{\mathrm{pi}}. \tag{2.16b}$$

For simplicity, bound electrons are assumed to only absorb N photons prior to removal from its parent nucleus. Above-threshold ionization, a phenomenon where an electron absorbs more than N photons, is also possible, but will be neglected here.

Photoionization, as well as any binary reaction that converts between a nucleus and an ion,

2.2 Electromagnetic effects

2.2.1 Maxwell's equation

Hydrocarbon materials such that one used in this project can be reasonably assumed to be isotropic and homogeneous. This simplifies the Maxwell's equations in matter because it allows two vectors that are difficult to compute, namely the polarization vector $\mathbf{P}$ and the magnetization vector $\mathbf{M}$, to be parallel to the electric and magnetic fields. Moreover, $\mu_r \approx 1$ is assumed since hydrocarbon are non-magnetic materials. This leads to the following form of Maxwell's equations in matter:

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon_r \epsilon_0} \tag{2.17a}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.17b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.17c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_f + \mu_0 \epsilon_r \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.17d}$$

In plasma, the free charge density ρ_f and current density $\mathbf{j}_f$ can be easily computed using electron and ion information:

$$\rho_f \equiv e(N_i - N_e) \tag{2.18a}$$

$$\mathbf{j}_f \equiv e(N_i \mathbf{u}_i - N_e \mathbf{u}_e) \tag{2.18b}$$

Though Equation 2.17 explicitly describes four vector equations, it can be easily shown that taking the divergence of Equations 2.17c and 2.17d will lead to Equations 2.17b and 2.17a, respectively. Therefore, only Equations 2.17c and 2.17d are necessary to model the $\mathbf{E}$ and $\mathbf{B}$ fields.

2.2.2 Lorentz force

The Lorentz force acts upon any charge particle has the form

$$\mathbf{f}_j = \pm e \left(\mathbf{E} + \mathbf{u}_j \times \mathbf{B} \right), \tag{2.19}$$

here the subscript j can be either e or i. Since a force is the rate of change of a momentum, Equation 2.19 can be expressed as

$$m_j \frac{dN_j \mathbf{u}_j}{dt} = \pm e N_j \left(\mathbf{E} + \mathbf{u}_j \times \mathbf{B} \right). \tag{2.20}$$

Since electrons are extremely lightweight, its acceleration due to the Lorentz force easily dominates any diffusive or gravitational effects, so they are safe to neglect. The case for ions is a little bit more nuanced; however, the Lorentz force is still expected to be the main momentum contributor.

2.2.3 Energy transport by electromagnetic waves

Electromagnetic waves themselves carry energy, and therefore capable of doing work on a charged system. According to Equation 2.20, the magnetic field does no work on charged particles as its force component is always perpendicular

to the charge velocity. The rate of work done by the electric field is then

$$\frac{\partial N_j e_j}{\partial t} = m_j \frac{dN_j \mathbf{u}_j}{dt} \cdot \mathbf{u}_j = \pm e N_j \mathbf{u}_j \cdot \mathbf{E}, \tag{2.21}$$

and should be equal to the rate of energy gained by the particles.

2.3 Reactions involving charged particles

While photoionization is expected to be the primary source of ionization, a couple of binary reactions can play a non-neglible role when it comes to mass balance.

Collisional ionization occurs when a free electron has enough kinetic energy

$$r_{\mathrm{ci}} = S_{\mathrm{ci}} N N_e \tag{2.22}$$

$$r_{\mathrm{rec}} = S_{\mathrm{rec}} N_i N_e \tag{2.23}$$

2.4 Collisions

Source terms (momentum):

$$\mathbf{p}_{en} = N N_e \sigma_{s,en} \|\mathbf{u}_e - \mathbf{u}\| \left(\mathbf{u}_e - \mathbf{u}\right) \tag{2.24}$$

$$\mathbf{p}_{in} = N N_i \sigma_{s,in} \|\mathbf{u}_i - \mathbf{u}\| \left(\mathbf{u}_i - \mathbf{u}\right) \tag{2.25}$$

2.5 Energy exchanges

Source terms (energy):

$$q'''_{ei} = \nu_{ei} N_e k_B (T_e - T_i) \tag{2.26}$$

$$q'''_{en} = \nu_{en} N_e k_B (T_e - T) \tag{2.27}$$

$$q'''_{in} = \nu_{in} N_i k_B (T_i - T) \tag{2.28}$$

2.6 Equations of state

Closure relations:

$$\rho = \rho(p, T) \tag{2.29}$$

$$N_e = N_e(p_e, T_e) \tag{2.30}$$

$$N_i = N_i(p_i, T_i) \tag{2.31}$$

Temperatures are obtained using equations of state:

$$E = E(p, T) \tag{2.32}$$

$$e_e = \frac{3}{2} k_B T_e \tag{2.33}$$

$$e_i = e_i(p_i, T_i) \tag{2.34}$$

2.7 The complete system of equations

The system of equations have a total of 21 nonlinear variables, organized into four categories. Five equations are needed to describe the fluid motion of neutral particles.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = \frac{M}{N_A}(\Gamma_{\text{rec}} - \Gamma_{\text{pi}} - \Gamma_{\text{ci}}) \tag{2.35}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \rho \mathbf{u} \otimes \mathbf{u} = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g} + \frac{M}{N_A}[\mathbf{f}_{en} + \mathbf{f}_{in} + \Gamma_{\text{rec}} \mathbf{u}_i - (\Gamma_{\text{pi}} + \Gamma_{\text{ci}})\mathbf{u}] \tag{2.36}$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot \rho \mathbf{u} H = -\nabla \cdot \mathbf{q}'' + \nabla \cdot [\mathbf{T} \cdot \mathbf{u}] + q'''_{\text{exc}} + q'''_{en} + q'''_{in} + \frac{M}{N_A}[\Gamma_{\text{rec}} E_i - (\Gamma_{\text{pi}} + \Gamma_{\text{ci}})E] \tag{2.37}$$

Electron motion requires another set of five equations.

$$\frac{\partial n_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e = \Gamma_{\text{pi}} + \Gamma_{\text{ci}} - \Gamma_{\text{rec}} \tag{2.38}$$

$$\frac{\partial n_e \mathbf{u}_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e \otimes \mathbf{u}_e = \frac{-\nabla p_e - e n_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B})}{m_e} - \mathbf{f}_{en} \tag{2.39}$$

$$\frac{\partial n_e \mathcal{E}_e}{\partial t} + \nabla \cdot n_e \mathbf{u}_e \mathcal{H}_e = q'''_{\text{pi}} - q'''_{en} - q'''_{ei} - e n_e \mathbf{u}_e \cdot \mathbf{E} \tag{2.40}$$

Ion motion is described by a very similar set of equations.

$$\frac{\partial n_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i = \Gamma_{\text{pi}} + \Gamma_{\text{ci}} - \Gamma_{\text{rec}} \tag{2.41}$$

$$\frac{\partial n_i \mathbf{u}_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i \otimes \mathbf{u}_i = \frac{-\nabla p_i + e n_i (\mathbf{E} + \mathbf{u}_i \times \mathbf{B})}{m_i} - \mathbf{f}_{in} + (\Gamma_{\text{pi}} + \Gamma_{\text{ci}})\mathbf{u} - \Gamma_{\text{rec}} \mathbf{u}_i \tag{2.42}$$

$$\frac{\partial n_i \mathcal{E}_i}{\partial t} + \nabla \cdot n_i \mathbf{u}_i \mathcal{H}_i = q'''_{ei} - q'''_{in} + e n_i \mathbf{u}_i \cdot \mathbf{E} + (\Gamma_{\text{pi}} + \Gamma_{\text{ci}})\mathcal{E} - \Gamma_{\text{rec}} \mathcal{E}_i \tag{2.43}$$

Finally, the electric and magnetic fields that arise from charge movement are taken from Maxwell's Equations.

$$\varepsilon_r \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - e(n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.44}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{2.45}$$

Chapter 3

Numerical methods

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